

Report Status FINAL

Route 2006 Ordered by:

Ironwood Cancer and Research Center Scottsdale8880 E Desert Cove Ave
Scottsdale, AZ 85260**Sonora Quest
Laboratories™**

A Subsidiary of Laboratory Sciences of Arizona

Rakesh Bagai, MD**Patient Information:****MILLER, ERIC J****Order #: 1505817030631872548 / NL120392006**

Account: 15058

Collected: 01/30/2026 12:56 PM

DOB: 07/22/1967 Age: 58Y-6M-8D**ID/MR#: 1759356**

Received: 01/31/2026 08:24 AM

Sex: M Fasting: No

Patient Lab ID:

Reported: 02/04/2026 01:41 PM

Patient Phone: 480-287-2227

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TEST	RESULTS	REFERENCE RANGES	UNITS	PL
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HEMATOLOGY**CBC w/ Differential, w/ Platelet**

WBC	8.7	4.0 - 11.0	k/mm3
RBC	4.73	4.30 - 6.00	m/mm3
Hemoglobin	15.0	13.5 - 17.0	g/dL
Hematocrit	43.7	40.0 - 53.0	%
MCV	92.4	78.0 - 100.0	fL
MCH	31.7	27.0 - 34.0	pg
MCHC	34.3	31.0 - 37.0	g/dL
Platelet Count	237	130 - 450	k/mm3
RDW(sd)	52.5 H	38.0 - 49.0	fL
RDW(cv)	15.5 H	11.0 - 15.0	%
MPV	10.3	9.0 - 12.0	fL
Segmented Neutrophils	63.4*		%
Lymphocytes	24.3		%
Monocytes	11.4		%
Eosinophils	0.0		%
Basophils	0.1		%
Absolute Neutrophil	5.5	1.5 - 7.8	k/uL
Absolute Lymphocyte	2.1	0.9 - 3.9	k/uL
Absolute Monocyte	1.0	0.2 - 1.0	k/uL
Absolute Eosinophil	0.0	0.0 - 0.6	k/uL
Absolute Basophil	0.0	0.0 - 0.2	k/uL
Immature Granulocytes	0.8		%
Absolute Immature Granulocytes	0.1	0.0 - 0.1	k/uL
NRBC RE, Nucleated Red Blood Cell	0.0	0.0 - 1.0	%

Percent

*Segmented

Automated Diff

Neutrophils:

CHEMISTRY

Protein, Total	6.2	6.0 - 7.7	g/dL
Beta-2-Microglobulin	1.8	0.8 - 2.4	mg/L

This test was performed using the Roche Cobas Tina-Quant B2MG immunoturbidometric assay.

Comprehensive Metabolic Panel

Glucose	97*	70 - 99	mg/dL
Urea Nitrogen (BUN)	16	7 - 28	mg/dL
Creatinine	0.75	0.68 - 1.37	mg/dL

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eGFRcr CKD-EPI	104*	≥60	mL/min/1.73m2
BUN/Creatinine Ratio	21.3	10.0 - 28.0	
Sodium	134 L	135 - 145	mmol/L
Potassium	3.8	3.6 - 5.3	mmol/L
Chloride	90 L	98 - 108	mmol/L
Carbon Dioxide (CO2)	31	20 - 31	mmol/L
Anion Gap	13	4 - 18	
Albumin	4.1	3.8 - 5.1	g/dL
Globulin	2.2	1.7 - 3.3	g/dL
Albumin/Globulin Ratio	1.9	1.3 - 2.7	
Calcium	9.6	8.7 - 10.4	mg/dL
Alkaline Phosphatase	59	40 - 140	IU/L
Alanine Aminotransferase	15	5 - 60	IU/L
Aspartate Aminotransferase	10 L	12 - 47	IU/L
Bilirubin, Total	0.3	≤1.2	mg/dL

*Glucose: Glucose reference range reflects fasting state.

*eGFRcr CKD-EPI: eGFRcr calculated using the CKD-EPI 2021 equation

Immunotyping (IT) w/ IgG, IgA, IgM Panel

IgG	786	694 - 1618	mg/dL
IgA	28 L	81 - 463	mg/dL
IgM	18 L	48 - 271	mg/dL
IT Interpretation	See Comment*		
*IT Interpretation:	Two monoclonal gammopathies, IgG, kappa type; duplication often represents the presence of a structural variant of an individual immunoglobulin clone.		

Protein Electrophoresis, Serum

Serum Proteins	Graph		
Albumin	3.61	3.52 - 5.29	g/dL
Alpha-1-Globulin	0.33	0.18 - 0.39	g/dL
Alpha-2-Globulin	0.89	0.45 - 0.94	g/dL
Beta Globulin 1	0.38	0.30 - 0.58	g/dL
Beta Globulin 2	0.69 H	0.20 - 0.52	g/dL
Gamma Globulin	0.30 L	0.70 - 1.50	g/dL
A/G Ratio	1.39		
M-Spike 1	0.63 H	≤0.00	g/dL
M-Spike 2	0.11 H	≤0.00	g/dL
PE Interpretation	See Comment*		
*PE Interpretation:	Monoclonal gammopathy in the beta region; compatible with plasma cell dyscrasia. Suggest serum immunotyping to verify findings and determine protein isotype.		

Monoclonal band co-migrates with the beta globulin. Quantitation is performed but may be affected. Recommend use of total immunoglobulin measurements (IgG, IgA, or IgM) to monitor changes.

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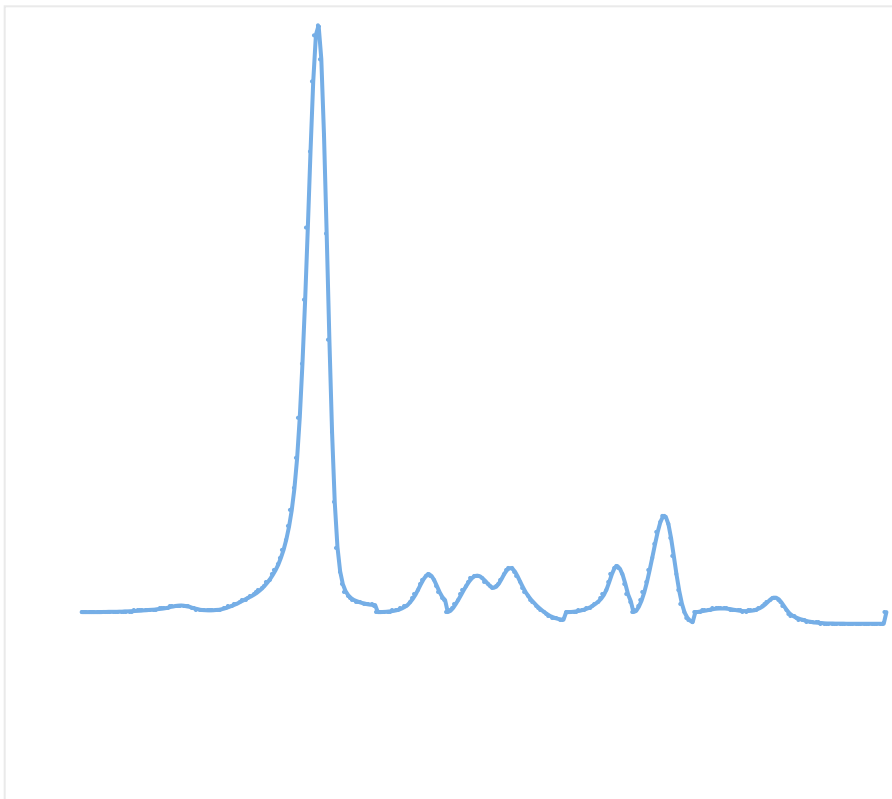
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Monoclonal gammopathy in the gamma region; compatible with plasma cell dyscrasia.
Suggest serum immunotyping to verify findings and determine protein isotype.

**Free Kappa/Lambda L/C Ratio, Serum****Free Kappa Light Chain****96.07 H**

3.30 - 19.40 mg/L

Free Lambda Light Chain

6.64

5.71 - 26.30 mg/L

Free Kappa/Lambda Ratio**14.47 H ***

0.26 - 1.65

***Free Kappa/Lambda
Ratio:**

Interpretation: In serum, the kappa/lambda ratio of whole immunoglobulin molecules is 2:1, whereas the kappa/lambda ratio of free light chains is 1:1.5. The latter is attributed to the occurrence of lambda light chains as dimers whose serum half-life is approximately 3 times longer than that of monomeric kappa light chains. Excess production of kappa or lambda light chains alters the kappa/lambda ratio. Alterations that fall outside of the normal range are attributed to the presence of monoclonal light chains. Monoclonal light chains are found in serum of patients with multiple myeloma, the light chain variant of MM, Waldenstrom's macroglobulinemia, mu-heavy chain disease, primary amyloidosis, light chain deposition disease, monoclonal gammopathy of undetermined significance, and lymphoproliferative disease such as B-CLL. Measurement of free light chain concentration in serum is useful for diagnosis, prognosis, monitoring disease activity and following response to therapy of these disorders.

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Page 3/4

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Chronic infection and chronic inflammatory diseases, as well as renal insufficiency, may be accompanied by a diffuse increase in both kappa and lambda free light chains, but the kappa/lambda ratio remains within the normal limits. The serum concentration of free light chains increases with age over 60 years; light chains may reach 50 mg/L in those 70-80 yrs of age; in these cases the kappa/lambda ratio still remains within normal limits.

Physicians, who are accustomed to the identification of clonal protein by electrophoretic means, may order immunofixation in addition to free light chain immunoassay. In rare instances, immunofixation may identify monoclonal light chain protein in the absence of abnormalities in the quantitative light chain immunoassays.

Katzman JA et al., Serum reference intervals and diagnostic ranges for free kappa and free lambda immunoglobulin light chains: Relative sensitivity for detection of monoclonal light chains. Clin Chem 2002;48:1437-1444.

Tests Ordered: CBC w/ Differential, w/ Platelet; Beta-2-Microglobulin; Comprehensive Metabolic Panel; Free Kappa/Lambda L/C Ratio,Serum; Monoclonal Gammopathy Monitoring Panel (SPE, IT)

Values Outside of Reference Range

TEST	RESULTS	REFERENCE RANGES	UNITS
RDW(sd)	52.5 H	38.0 - 49.0	fL
RDW(cv)	15.5 H	11.0 - 15.0	%
Sodium	134 L	135 - 145	mmol/L
Chloride	90 L	98 - 108	mmol/L
Aspartate Aminotransferase	10 L	12 - 47	IU/L
IgA	28 L	81 - 463	mg/dL
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M-Spike 1	0.63 H	≤0.00	g/dL
M-Spike 2	0.11 H	≤0.00	g/dL
Free Kappa Light Chain	96.07 H	3.30 - 19.40	mg/L
Free Kappa/Lambda Ratio	14.47 H	0.26 - 1.65	

Values listed above may not include all results considered abnormal for this patient (e.g., text-only results, such as those for some pathology/cytology specimens, and results for analytes without established reference ranges will not appear). Always review the entire patient report and correlate all results with the patient's clinical condition.

Unless otherwise noted, testing performed by: Sonora Quest Laboratories, 424 S 56th St, Phoenix, AZ 85034
800.766.6721 Laboratory Director: Dr. Timothy McDonald

End of Report**MILLER, ERIC J Order #: 1505817030631872548 / NL120392006 - FINAL Report**

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